Crude oil trades hedging with targeting volatility by using Large Language Models

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This paper is devoted to the crude oil trades hedging with targeting volatility by using Large Language Models (LLM), comparing efficiency and profitability of LLaMA2 and GPT3.5.

Empirical studies of crude oil data provide strong evidence for price jumps (Larsson and Nossman (2011) and Askari and Krichene (2008)). There are different approaches for hedging derivative securities in an incomplete market accommodating price jumps. An interesting practical question is whether quadratic hedging strategies are able to outperform some of the more traditional hedging strategies used in practice, such as the Black-Scholes delta or the Merton delta hedge (accounting for the jumps) [1]. Poulsen, Schenk-Hoppe and Ewald (2009) conducted an empirical study on the performance of quadratic hedging strategies using share-index data.

One important feature of crude oil markets is that prices exhibit jumps, hence we consider the situation where the market incompleteness arises from jumps in crude oil prices. The locally risk-minimizing hedging strategies we derive are different from the ones obtained in Poulsen, Schenk-Hoppe and Ewald (2009) and so is the calibration and dynamic updating. However, to examine the empirical performance of the hedging strategies, we can follow the same methodology as used in Poulsen, Schenk-Hoppe and Ewald (2009) [2]. The empirical performance of locally risk-minimizing hedging strategies is examined using NYMEX crude oil futures and options data. We observe that locally risk-minimizing hedges derived from a properly calibrated jump-diffusion model significantly outperform the Black-Scholes delta hedges and the Merton delta hedges. The significance of the improvement in the hedge performance by locally risk-minimizing hedges is of similar magnitude as in Poulsen, Schenk-Hoppe and Ewald (2009).

Previous studies diversify their results in the shape of the relationship between oil price uncertainty and investment. For instance, studies such as Mohn & Misund (2009) and Elder & Serietis (2020) find a linear relationship between oil price uncertainty and investment; while Henriques & Sadorsky (2021) found a *U* curve relationship of them. Therefore the variable of oil price volatility square will be introduced as an optional variable. If the specification of model with the variable shows a better result than the one without it, then the shape of the relationship would be a curve line.

Volatility targeting is a portfolio management tool aimed at managing commodities portfolio risk by targeting a particular volatility level and adjusting the positioning of the portfolio in an attempt to stay close to the volatility target. Although it is possible to implement volatility targeting with any asset, portfolios of derivatives provide a cost-effective implementation as trading is often done on a daily basis. Following the research of Moreira and Muir (2017) the idea is very straightforward; one needs to simply scale asset weights in the portfolio by the ratio of the target volatility and the expectation of future volatility given the current asset mix. Obviously, the value of the targeting is greatly impacted by the quality of the volatility expectation model, but since these estimates are over short horizons, the fact that volatility tends to cluster makes this an “easier” effort. In equation form, the return of the volatility managed commodity portfolio, is the volatility scaled return of the unmanaged portfolio which is simply the sum product of the asset weights and their returns.

Crude oil is a pivotal global commodity whose price fluctuations permeate the broader economy through channels such as transportation costs, energy-weighted inflation indices, and corporate procurement and inventory strategies. Accurate forecasting of oil prices, even for single-step daily predictions, holds significant practical value. Marginal but consistent accuracy gains around price turning points can substantially mitigate risks associated with supply disruptions and excessive hedging costs.

Such buying and selling of futures by inventory hedgers driven by the shape of the futures curve is rather mechanical and somewhat predictable. The cost of carry, which includes transportation, logistics, borrowing costs and the convenience yield of holding inventories depend on the economics of individual traders, which makes hedging pressure vary with the steepness of the curve. In general,

the steeper the contango, the more widespread hedging becomes among inventory managers, which increases the selling pressure in the futures market.

One pivotal factor that threads through virtually every aspect of oil trading is the shape of the futures curve. It arises everywhere while often hiding under different names in specific applications. Some economists may recognize it as the own rate of interest, fundamental traders know it as the convenience yield, and financial investors call it the roll yield. Systematic traders refer to the same term by yet another name, the commodity carry.

The business model of the storage trader is to generate low-risk steady returns by providing the service of storing oil. In such a business model, the risk of volatile oil prices must be eliminated by hedging in the futures market. As soon as a physical barrel is purchased for storage, a financial barrel is sold in the futures market to reduce the risk. This downward pressure on futures price occurs when the contango is steep enough to cover the cost of storage. If the market flips to backwardation or the contango becomes too narrow to economically hold inventories, then the storage hedger is incentivized to pull oil out of storage and buy back short futures hedges. This creates an upward pressure on the futures price.

Oil trades hedging in Incomplete Markets

In the context of quadratic hedging, there are two approaches to define the mean-square criterion to be minimized. In the first approach, the hedging strategy is supposed to be self-financing, and one minimizes the quadratic hedging error at maturity. This is to minimize the expected squared difference between the terminal value of the hedging portfolio $V_T(\varphi)$ and the option payoff H:

$$\inf_{V_0,\varphi} \mathrm{E}[|V_T(\varphi) - H|^2] ,$$

subject to:

$$V_T(\varphi) = V_0 + \int_0^T \varphi_t^0 dB_t + \int_0^T \varphi_t^1 dS_t . \qquad (1)$$

where:

B_t and S_t are the prices of a risk-free asset and a risky asset, respectively. When the interest rate is constant, say r, we choose the zero-coupon bond with maturity T to be the risk-free asset. In this case, $B_t = e^{-r(T-t)}$;

φ_t^0 and φ_t^1 are the numbers of units of the risk-free asset and the risky asset held in the hedging portfolio at time t.

After discounting with the factor B_t, the corresponding hedging problem can be written as:

$$\inf_{V_0,\varphi} \mathbf{E}[|\hat{V}_T(\varphi) - \hat{H}|^2],$$

subject to:

$$\hat{V}_T(\varphi) = \hat{V}_0 + \int_0^T \varphi_t^1 d\hat{S}_t . \qquad (2)$$

Here $\hat{V}_t := B_t^{-1} V_t$ and $\hat{S}_t := B_t^{-1} S_t$. In the literature this approach is referred to as mean-variance hedging.

When strategies are not self-financing they are required to replicate the option's payoff exactly, i.e., $V_T(\varphi) = H$. In an incomplete market, this means that the option's seller has to continuously inject money to, or withdraw money from the hedging portfolio. The cumulative process of the amounts of funds injected, or withdrawn, is called the cost process and is given as follows:

$$C_t(\varphi) = V_t(\varphi) - G_t(\varphi) \qquad (3)$$

where

$$V_t(\varphi) = \varphi_t^0 B_t + \varphi_t^1 S_t \qquad (4)$$

and the (self-financing) gains process is given by

$$G_t = \int_0^t \varphi_s^0 dB_s + \int_0^t \varphi_s^1 dS_s. \qquad (5)$$

The discounted cost process is then given by

$$\hat{C}_t = \varphi_t^0 + \varphi_t^1 \hat{S}_t - \int_0^t \qquad (6)$$

We start with a complete, filtered probability space $(\Omega, F, \{F_t\}, P)$, where $\{F_t\}$ is a filtration satisfying the usual conditions, (i.e. right continuity and P-completeness) and P is the subjective probability measure. We suppose that the probability space (Ω, F, P) is rich enough to describe uncertainties arising from normal fluctuations and jumps in the price of the underlying risky asset, i.e. the crude oil. We assume that prices S_t of the crude oil are governed by the following Merton-type jump-diffusion model.

Under the above assumption, Zhang (1994) showed that a minimal martingale measure exists, if and only if the following condition is satisfied:

$$-1 \le \eta = \frac{\mu + \lambda m - r}{\lambda(\delta^2 + m^2)} \le 0. \quad \sigma^2 +$$

This condition will be relevant for computing the locally risk minimizing hedge. If it is satisfied, the minimal martingale measure Q^{min} is determined by the following Radon-Nikodym derivative:

$$\frac{dQ^{min}}{dP} = \exp\left(-\sigma\eta W_T + \lambda\eta m T - \frac{\sigma^2\eta^2}{2} T\right) \prod_{j=1}^{N_t} (1 - \eta Y_j)^{N_T}. \qquad (7)$$

The price process of the crude oil under the risk-neutral measure Q^{min} can then be expressed as

$$\frac{dS_t}{S_{t-}} = rdt + dU_t,$$
$$U_t = \lambda[\eta(m^2 + \delta^2) - m]t + \sigma W_t' + \sum_{i=1}^{N_t'} \Delta U_i, \tag{8}$$

where under Q

- W_t' is a standard Brownian motion with respect $\{F_t\}$;
- N_t' is a Poisson process with constant intensity $\lambda' = \lambda(1 - \eta m)$;
- the jump sizes ΔU_i are i.i.d. with common distribution F_u, where

$$dF_u(x) = \frac{1 - \eta x}{1 - \eta m} dF(x) .$$

We shall use Q^{min} as the pricing measure for computing the locally risk-minimizing hedging strategy. One advantage of this pricing measure is that it can price explicitly the jump risk. This is in contrast with other risk-neutral measures which remove the jump component entirely (see Benth (2003), Bouleau (1998), Föllmer and Schweizer(1990) and (1991)).

Computing the Locally Risk Minimizing Hedge by using LLM models LLaMA2 and GPT3.5.

In this section we apply results from Tankov (2022) and Cont and Tankov (2024) in order to compute the mean-variance and locally risk-minimizing hedges in the jump diffusion framework presented in the last subsection.

Combining a Brownian motion with drift and a compound Poisson process, we obtain the simplest case of a jump diffusion - a process which sometimes jumps and otherwise has a continuous but random evolution between the jump times.

$$X_t = \gamma t + \sigma W_t + \sum_{i=1}^{N_T} + Y_i \,. \qquad (6)$$

The best known model of this type in finance is the Merton model under which the discounted price process of the risky asset is given by

$$\hat{S}_t = \hat{S}_0 e^{Xt} \,, \qquad (7)$$

where the process Xt is defined as above with γ = μ − r and the jump sizes Yi have a common Gaussian distribution. This fits well into the class of models discussed in the previous section.
Indeed, a more general type of Levy process can be represented in the form

$$X_t = \gamma t + \sigma W_t + Z_t \,, \qquad (8)$$

where Z_t is a jump process with possibly infinitely many jumps.
Under a risk neutral probability, the process e^{Xt} , interpreted as discounted asset price, must be a martingale. From the Lèvy-Khintchine formula we obtain by choosing $u = i$, that a necessary and sufficient condition for e^{Xt} to be a martingale under P and let's denote by g its log payoff function, i.e. $H(e^x) \equiv g(x)$.

$$\gamma + \frac{\sigma^2}{2} + \int_{\mathrm{R}} (e^y - 1 - y1_{|y|\le 1})v(dy) = 0$$

Suppose that under a given risk neutral measure Q there exists $R \neq 0$ such that

$$g(x)e^{-Rx} \quad \text{has finite variation on } R \tag{9}$$

$$g(x)e^{-Rx} \in L^1(\mathrm{R}) \tag{10}$$

$$E^{\mathrm{Q}}[e^{RX_{T-t}}] < \infty \quad \text{and} \quad \int_{\mathrm{R}} \frac{|\psi_{T-t}(u - iR)|}{1 + |u|} du < \infty, \tag{11}$$

where ψ_{T-t} denotes the characteristic function of X_{T-t} under Q. Then the price at time t of the European option with payoff function H computed under Q can be represented as

$$C(t, S_t) = e^{-r(T-t)}E^{\mathrm{Q}}[H(S_T)|\mathrm{F}_t] \tag{12}$$

The Heston Model as the data generating process for crude oil trades hedging

The Heston model features the dynamics

$$dS_t = S_t\,\mu dt + \sqrt{V_t}\sqrt{1-\rho_t^2}dW^1 + {}_t\rho dW^2$$
$$dV_t = \kappa(\theta - V_t)dt + \rho\sqrt{V_t}dW^2. \qquad (13)$$

Our simulation experiments use the parameter estimates obtained from the comprehensive empirical study conducted in Krichene (2008) which can be seen as reflecting a consensus in the literature. We assume that the Heston model is the "true" data-generating process and look at hedgers who adopt the Merton model or the Black-Scholes model [3]. This investigation of the performance of the locally risk-minimizing delta hedge, the delta hedge in the (wrong) Merton model and the delta hedge in the (wrong) Black-Scholes model is carried out as follows:

- simulate stock prices and volatilities from the Heston model;
- for each path implement the Merton-based locally risk-minimizing strategy (using the initially calibrated parameters and the simulated Merton-sense local variance along each path), as well as the two delta hedges;
- evaluate the performance criterion below.

Application of LLM for hedging error minimization

Key principles of LLM (LLaMA2 and GPT3.5) application are the next

- Design an LLM-based return prediction model (LLaMA2) comprising the volatility targeting and forecasting modules;
- The basic hypothesis is that the crude oil hedging and volatility targeting representations from encoder-only and decoder-only LLMs perform differently due to their distinct methods of encoding sequences in pre-training and fine-tuning. Thus, I proposed to compare GPT3.5 and decoder-only LLMs (LLaMA2) as the representation module of delta hedging model (for commodity derivatives);
- As LLM-generated oil trade hedge and volatility targeting representations are at the token level, I presented two simple methods to integrate token representations into the forecasting module: bottleneck and aggregated representations;

Simulation of Crude oil trades hedging by using LLaMA2 and GPT3.5

We now need to simulate crude oil trades hedging process by using the real world as the data generating process. By doing this we evade any critique of model choice. If (as we demonstrate) our strategies perform better in real life trading than alternative, classically used, strategies, then it does not matter under which model assumptions applied investment strategies have been obtained. The improvement in hedging error in real life trading is the only thing that eventually matters to the practitioner.

At a point in time t the hedger sets up a position that holds φ_t units of the underlying S_t and $\varphi^0 = e^{-rt}(V_t - \varphi_t S_t)_{\,t}$ in a market account, where V_t denotes the observed market price of the oil

(commodity) option to be hedged. This position then requires an investment of V_t[4]. We consider the locally risk-minimizing hedge obtained earlier, the Merton delta hedge, and as a benchmark, we also include a simple Black-Scholes delta hedge, i.e. the hedger holds a number of units of the underlying that is equal to the Black-Scholes delta. We assume that at a given date *t*, the hedger calibrates his model's parameters to the option prices observed at this date, using the regularized calibration approach. The idea and design of this analysis are similar to those in Poulsen, Schenk-Hoppe and Ewald (2009), and they are described as follows: the different hedging strategies are applied to the actual price paths of the real data and the path-specific terminal cost associated with each is recorded. This is repeated over and over again using options and futures market data in order to compute the hedge error which is defined to be

$$\text{Hedge error} = 100 \times \frac{\sqrt{Var^{P}}\, C_T}{e^{-rT}\,\mathrm{E}^{\min}([S_T - K]^{+})}. \qquad (13)$$

That is the standard deviation of the cost process at expiry divided by the initial cost of the option value in percentage terms. This error measurement tracks the hedge all the way until the date of expiry; smaller hedge errors indicate less variance associated with the final cost of the trading strategy. The same measurement of the hedge error was applied in Poulsen, Schenk-Hoppe and Ewald (2009). For completeness and full comparison, we now also include the Poulsen, Schenk-Hoppe and Ewald (2009) Heston locally risk minimizing hedge into our analysis.

The locally risk-minimizing hedges (both Merton and Heston) lead to a smaller hedge error than the ordinary Merton and Black-Scholes delta hedges.

Heston data generating process **Chart 1**

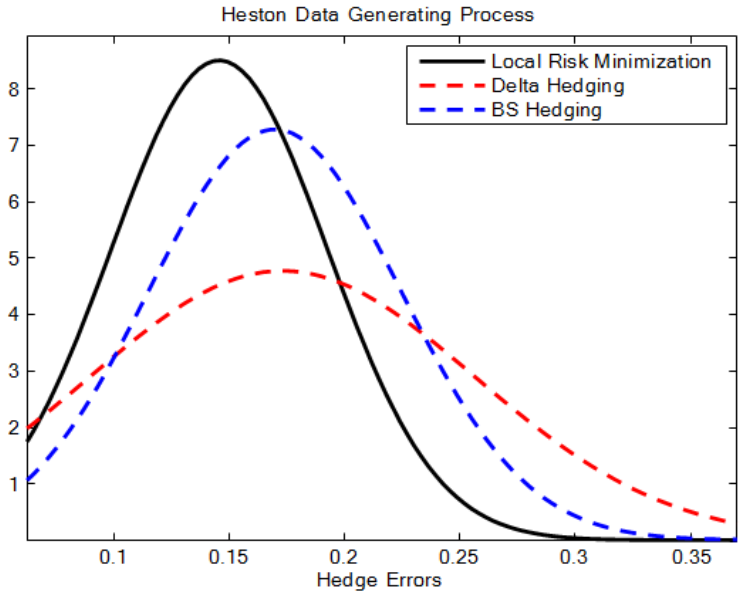


Table 1

Hedge errors in misspecified setup with Heston data generating process. The target option is a 1 year ATM put option on crude oil

Method		Hedge minimizing	Locally-risk Merton Black-Scholes (Delta)
LLaMA2	Hedge error	3.81	6.32
GPT3.5	Hedge error	4.96	8.69
Without using LLM	Hedge error	2.96	12.14

As we can see from the Table 1, application of LLaMA2 and GPT 3.5 significantly reduce hedge error of oil derivatives or physical (naked) oil position, given that LLaMA2 outperforms GPT 3.5 regarding hedge error.

Validation procedure for LLaMA2 and GPT 3.5 which are used for Momentum Strategy with Volatility targeting Performance

Another important procedure of LLaMA2 and GPT3.5 is validation performed via *VALTEST*, a framework for automatically validating test cases generated by Large Language Models (LLMs). The goal of *VALTEST* is to improve the reliability of LLM-generated test cases by leveraging token probabilities to predict the validity of test cases [3]. The framework evaluates the validity and coverage metrics for test cases generated from LLaMA2 and GPT3.5.

Key Components of VALTEST

1) *Feature extraction.* The framework extracts statistical features from the token probabilities for both function inputs and expected outputs. These features include *mean*, *max*, *min*, *sum*, *variance*, and total token counts for both the top-predicted token and the second-predicted token;
2) *Model training and Evaluation.* The machine learning models (e.g., logistic regression, random forest, ensemble models) are trained using the extracted features. The models predict the validity of the test cases by evaluating their probability distributions. A *K*-fold cross-validation is used for training and testing across different LLM-generated test suites;
3) *Test Case Generation.* Test cases are generated by prompting LLMs using function signatures and task descriptions from datasets such as *HumanEval*, *MBPP*, and *LeetCode*. Token probabilities are captured during this process to extract features for model training;
4) *Test Case Curation.* Invalid test cases are identified and corrected through a chain-of-thought reasoning process. This step ensures that test cases reflect the expected behavior of the functions, even if the LLM initially generated invalid assertions.

Conclusion:

This work conducts an empirical study to extensively investigate the effectiveness and efficiency of crude oil trading hedging by using LLM-based techniques (LLaMA2) and GPT3.5. LLaMA2 significantly surpass GPT3.5 results, with the fine-tuned code embedding hedging being the most effective.

Key inferences:

- We calculated quadratic (mean-variance and locally risk-minimizing) hedging strategies for a class of jump diffusion models. We compared the performance of locally risk minimizing strategies with delta hedging strategies à la Merton and Black/Scholes. Delta hedging strategies for commodity (oil) trading only consist of a diffusion component and thus hedge diffusion risk but do not hedge jump risk;
- This research conducts an empirical study to extensively investigate the effectiveness and efficiency of crude oil trading hedging by using LLM-based techniques (LLaMA2) significantly surpass GPT3.5 results, with the fine-tuned code embedding hedging being the most effective;
- LLM-based techniques (LLaMA2) significantly surpass GPT3.5 results regarding hedging error for oil trades (commodity derivatives and physical oil contracts). Moreover, LLM-based techniques strike an excellent balance between crude oil trading costs and effectiveness. My work also paves the way for promising future research such as the study of efficiency of other crude oil trading strategies, combining and reinforced with other techniques.
- Our empirical tests using energy market data showed that locally risk-minimizing hedges offer major improvements over standard delta hedges: locally risk minimizing hedges are easily computable and one is never worse off when using them, but often better off by quite a margin. I found that locally risk minimizing hedge strategies are less sensitive to stock price changes than delta hedging strategies. Moreover, if the expected excess return is positive and the expected jump size is significantly different from zero and positive (negative) then the locally risk minimizing ratios are substantially larger (smaller) than the delta hedge ratios for out of the money (in-the-money) calls.

References:

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